The Regularity Lemma for Stable Graphs and its Applications in Property Testing

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Table of contents

Introduction to Szemerédi's Regularity Lemma

The idea of the SzRL is that all graphs can be partitioned in such a way that most pairs of parts behave in a quasi-random way, a property which is formalized as *regularity*.

Definition

Given $\epsilon > 0$ and a graph G, a pair of (not necessarily disjoint) subsets of vertices $A, B \subseteq G$ is said to be ϵ-*regular* if for all $A' \subseteq A$ and $B' \subseteq B$ such that $|A'| \geq \epsilon|A|$ and $|B'| \geq \epsilon|B|$, we have

$$|d(A', B') - d(A, B)| \leq \epsilon.$$

When most of the pairs satisfy this property, we say that the partition is *regular*.

Definition

Let G be a graph and let $\epsilon > 0$. An *ϵ-regular partition* of G is a partition of its vertex set into k parts $\{A_1, \ldots, A_k\}$ with remainder set B such that:

- $|B| \leq \epsilon|G|$, and may be empty.
- All but at most ϵk^2 of the pairs (A_i, A_j) with $1 \leq i < j \leq k$ are ϵ-regular.

Theorem (Szemerédi's Regularity Lemma, [?])

For every $\epsilon > 0$ and every positive integer m, there exists a positive integer $M = M(\epsilon, m)$ such that every graph with at least m vertices admits an ϵ-regular partition $\{A_1, \ldots, A_k\}$ and remainder B with $m \leq k \leq M$, with $|A_1| = \cdots = |A_k|$.

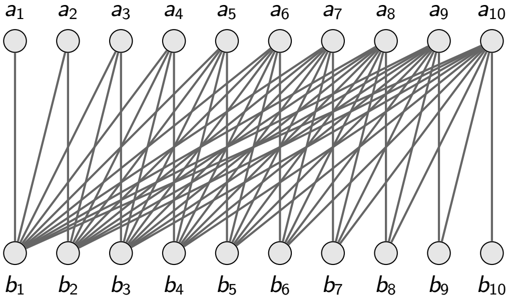
Crucially, the theorem proves that the number of parts k does not grow with the size of the graph, but only depends on ϵ (and m).

However, the lemma has two major drawbacks:

1. The bound on the number of parts M is a *tower-type* function of ϵ, i.e. a power-tower of height polynomial in $1/\epsilon$. This is unavoidable, as shown by Gowers [**?**].
2. The partition may contain some irregular pairs. This is also unavoidable, and the half-graph H_n is an example.

Stable Graphs and the Stable Regularity Lemma

Stable graphs are those that do not contain a *half-graph* as a bi-induced subgraph.

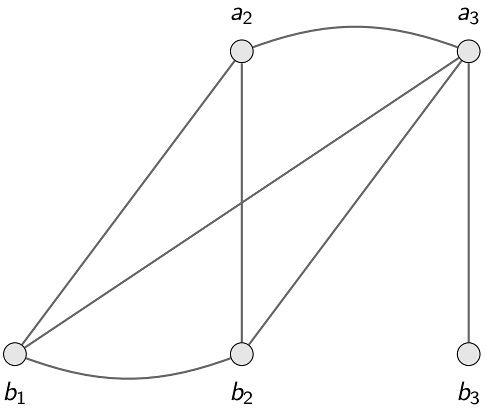


	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8	b_9	b_{10}
a_1	1	0	0	0	0	0	0	0	0	0
a_2	1	1	0	0	0	0	0	0	0	0
a_3	1	1	1	0	0	0	0	0	0	0
a_4	1	1	1	1	0	0	0	0	0	0
a_5	1	1	1	1	1	0	0	0	0	0
a_6	1	1	1	1	1	1	0	0	0	0
a_7	1	1	1	1	1	1	1	0	0	0
a_8	1	1	1	1	1	1	1	1	0	0
a_9	1	1	1	1	1	1	1	1	1	0
a_{10}	1	1	1	1	1	1	1	1	1	1

Definition

Let G be a graph. We say that G has the *k-order property* if there exist two sequences of vertices $\langle a_i \mid i \in \{1, \ldots, k\}\rangle$ and $\langle b_i \mid i \in \{1, \ldots, k\}\rangle$ such that for all $i, j \leq k$, $a_i R b_j$ if and only if $i \geq j$. Otherwise, we say that G has the *non-k-order property* or that G is *k-stable*.

The two sequences need not to be disjoint, and any combination of edges and non-edges may appear within the vertices of each sequence.



	a_2	a_3	b_1	b_2	b_3
a_2	0	1	1	1	0
a_3	1	0	1	1	1
b_1	1	1	0	1	0
b_2	1	1	1	0	0
b_3	0	1	0	0	0

Theorem (Stable Regularity Lemma, [?])

Each k-stable graph G admits an equitable partition of its vertex set with at most $c(k) \cdot (1/\epsilon)^{2^{k+1}}$ parts such that ALL pairs of parts are ϵ-regular and have density either less than ϵ or greater than $1 - \epsilon$.

Notably both limitations of the SzRL are overcome: the bound on the number of parts is polynomial in $1/\epsilon$, and there are no irregular pairs.

Sketch of the Proof of the Stable Regularity Lemma

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Property Testing of H-freeness in Stable Graphs

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Conclusion and Future Work

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Thank you!

You can now rate the presentation. Was it *good*, *excellent* or *exceptional*?